Factor Analysis

Readings: Johnson & Wichern, Chapter 9;
Zelterman, Section 8.5; Izenman, Section 15.4

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Agenda

Background

Factor Model Analysis

Stock Price Data Example

Factor Analysis: The Big Picture

Goal: Can correlations among many observed variables be explained by a few **latent factors**?

Model for an observed variable

$$X_j - \mu_j = \underbrace{\sum_{k=1}^{m} \ell_{jk} F_k}_{\text{Common factors}} + \underbrace{\varepsilon_j}_{\text{Specific variation}}$$

- **Common factors** explain variation shared across variables.
- **Specific variation** is unique to each variable.

Key Idea

Observed variables are correlated because they depend on shared latent factors.

Motivating Example: Intelligence Test Scores [Smith & Stanley (1983)]

Six intelligence tests were administered to 112 individuals: **general, picture, blocks, maze, reading,** and **vocabulary**.

The sample correlation matrix is:

	general	picture	blocks	maze	reading	vocab
general	1.000	0.466	0.552	0.340	0.576	0.514
picture	0.466	1.000	0.572	0.193	0.263	0.239
blocks	0.552	0.572	1.000	0.445	0.354	0.356
maze	0.340	0.193	0.445	1.000	0.184	0.219
reading	0.576	0.263	0.354	0.184	1.000	0.791
vocab	0.514	0.239	0.356	0.219	0.791	1.000

Key Question: Can the correlations among these six tests be explained by just one or two **underlying (latent) factors**?

Factor Analysis: Modeling Covariance with Latent Factors

Factor Analysis (FA) models the covariance among p observed variables $\boldsymbol{X} = (X_1, \ldots, X_p)^T$ using a smaller set of $m < p$ unobserved **latent factors** $\boldsymbol{F} = (F_1, \ldots, F_m)^T$.

Three main objectives:

1. **Data reduction:** Summarize the covariance structure of p variables using $m < p$ common factors.

2. **Data interpretation:** Identify latent factors underlying the observed correlations (**exploratory FA**).

3. **Theory testing:** Assess whether a hypothesized factor structure is consistent with the data (**confirmatory FA**).

PCA vs. Factor Analysis: Key Differences

	PCA	**Factor Analysis**
Goal	Summarize total variation	Explain covariance structure
Variables	Components are linear combinations of $\boldsymbol{X}$	Factors are latent (unobserved) variables
Model	No latent-variable model	Latent-factor model
Variance	Uses total variance	Separates common and unique variance
Key object	Eigenvectors	Factor loadings

Key Distinction

PCA: How can we summarize $\boldsymbol{X}$?

FA: What latent structure may explain $\boldsymbol{X}$?

Factor Analysis: The Statistical Model

Let $\boldsymbol{X}$ be a p-dimensional random vector with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$.

The factor model expresses each observed variable using $m < p$ **common latent factors**:

$$
\begin{aligned}
X_1 &= \mu_1 + \ell_{11} F_1 + \cdots + \ell_{1m} F_m + \varepsilon_1, \\
X_2 &= \mu_2 + \ell_{21} F_1 + \cdots + \ell_{2m} F_m + \varepsilon_2, \\
&\vdots \\
X_p &= \mu_p + \ell_{p1} F_1 + \cdots + \ell_{pm} F_m + \varepsilon_p.
\end{aligned}
$$

Model components:

- ℓ_{jk}: factor loading of variable j on factor k.
- F_k: common latent factor shared across observed variables.
- ε_j: specific error associated with variable j.

Factor Model: Matrix Form and Identification

The factor model can be written compactly as

$$\boxed{\boldsymbol{X} = \boldsymbol{\mu} + \boldsymbol{L}\boldsymbol{F} + \boldsymbol{\varepsilon}}$$

where

- $\boldsymbol{L} = (\ell_{jk})_{p\times m}$: factor loading matrix.
- $\boldsymbol{F} = (F_1, \ldots, F_m)^T$: vector of common latent factors.
- $\boldsymbol{\varepsilon} = (\varepsilon_1, \ldots, \varepsilon_p)^T$: vector of specific errors.

Key Challenge

Unlike ordinary regression, the factors $\boldsymbol{F}$ are **unobserved**.

We therefore need assumptions to identify and estimate the factor model.

Factor Model: Assumptions and Covariance Structure

Assumptions:

1. **Common factors:**

$$\mathbb{E}(\boldsymbol{F}) = \mathbf{0}, \qquad \mathrm{Var}(\boldsymbol{F}) = \boldsymbol{I}_m$$

   Factors are standardized and mutually uncorrelated.

2. **Specific errors:**

$$\mathbb{E}(\boldsymbol{\varepsilon}) = \mathbf{0}, \qquad \mathrm{Var}(\boldsymbol{\varepsilon}) = \boldsymbol{\Psi} = \mathrm{diag}(\psi_1, \ldots, \psi_p)$$

   Errors are mutually uncorrelated.

3. **Factors and errors:**

$$\mathrm{Cov}(\boldsymbol{F}, \boldsymbol{\varepsilon}) = \mathbf{0}$$

   Common factors and specific errors are uncorrelated.

Resulting Covariance Structure

$$\boxed{\boldsymbol{\Sigma} = \boldsymbol{L}\boldsymbol{L}^T + \boldsymbol{\Psi}}$$

Factor Models: Communality, Uniqueness, and Covariance

From $\boldsymbol{\Sigma} = \boldsymbol{L}\boldsymbol{L}^T + \boldsymbol{\Psi}$, the variance of each observed variable is

$$\mathrm{Var}(X_j) = \underbrace{h_j^2}_{\text{Communality}} + \underbrace{\psi_j}_{\text{Uniqueness}}$$

- **Communality:** Variance explained by the common factors,

$$h_j^2 = \sum_{k=1}^{m} \ell_{jk}^2.$$

- **Uniqueness:** Variance specific to variable X_j,

$$\psi_j = \mathrm{Var}(\varepsilon_j) = \mathrm{Var}(X_j) - h_j^2.$$

Covariance between distinct variables ($i \neq j$):

$$\mathrm{Cov}(X_i, X_j) = \sum_{k=1}^{m} \ell_{ik}\ell_{jk}.$$

Why Do We Want $m \ll p$ Common Factors?

Goal: Explain the covariance structure of p observed variables using only m common factors.

- An unrestricted covariance matrix $\boldsymbol{\Sigma}$ has
$$\frac{p(p+1)}{2}$$
distinct variance and covariance parameters.
- A factor model with m factors uses pm loadings and p specific variances:
$$pm + p = p(m+1).$$

Example: $p = 12$, $m = 2$

$$\underbrace{\frac{12(13)}{2} = 78}_{\text{Covariance entries}} \quad \longrightarrow \quad \underbrace{12(2+1) = 36}_{\text{Factor-model parameters}}$$

How Many Common Factors Should We Retain?

Choosing m requires balancing **model fit, parsimony, and interpretability**.

1. **Scree plot / eigenvalues:** Look for a sharp drop or leveling off in eigenvalues.
2. **Parallel analysis:** Compare observed eigenvalues with those expected from random data.
3. **Likelihood-ratio test:** Assess model fit under maximum likelihood factor analysis.
4. **Interpretability:** Retain factors with meaningful loading patterns.

Practical Takeaway

Do not select m based on a single diagnostic alone.

Estimating the Factor Model

Given the number of factors m, estimate $\boldsymbol{L}$ and $\boldsymbol{\Psi}$ in

$$\boldsymbol{\Sigma} = \boldsymbol{L}\boldsymbol{L}^T + \boldsymbol{\Psi}.$$

1. Principal Component Method

- Uses the leading eigenvalues and eigenvectors of the sample covariance (or correlation) matrix.
- Initial loading estimates:

$$\hat{\boldsymbol{L}} = (\sqrt{\hat{\lambda}_1}\hat{\boldsymbol{e}}_1, \ldots, \sqrt{\hat{\lambda}_m}\hat{\boldsymbol{e}}_m).$$

- Estimate specific variances from the remaining diagonal variation.

2. Maximum Likelihood (ML)

- Assumes $\boldsymbol{X}_i \overset{\text{iid}}{\sim} N_p(\boldsymbol{\mu}, \boldsymbol{L}\boldsymbol{L}^T + \boldsymbol{\Psi})$.
- Estimates $\boldsymbol{L}$ and $\boldsymbol{\Psi}$ jointly by maximizing the likelihood.
- Supports likelihood-based goodness-of-fit tests.

Goodness-of-Fit Test for Factor Analysis

Question: Can m common factors adequately explain the observed covariance structure?

$$H_{0(m)}: \quad \boldsymbol{\Sigma} = \boldsymbol{L}\boldsymbol{L}^T + \boldsymbol{\Psi},$$
$$H_1: \quad \boldsymbol{\Sigma} \text{ is unrestricted.}$$

Bartlett-corrected likelihood-ratio test:

$$T_m = \left(n - 1 - \frac{2p + 4m + 5}{6}\right) \log\left(\frac{|\hat{\boldsymbol{L}}\hat{\boldsymbol{L}}^T + \hat{\boldsymbol{\Psi}}|}{|\hat{\boldsymbol{\Sigma}}|}\right).$$

Under $H_{0(m)}$, approximately

$$T_m \sim \chi^2_\nu, \qquad \nu = \frac{(p-m)^2 - p - m}{2}.$$

- **Small p-value:** Evidence that m factors are insufficient.
- **Large p-value:** No significant evidence of lack of fit.

Scale Equivariance of Factor Analysis

Suppose the observed variables are rescaled:

$$\boldsymbol{Y} = \boldsymbol{C}\boldsymbol{X}, \qquad \boldsymbol{C} = \mathrm{diag}(c_1, \ldots, c_p),$$

where each $c_j \neq 0$.

Since $\boldsymbol{\Sigma} = \boldsymbol{L}\boldsymbol{L}^T + \boldsymbol{\Psi}$,

$$\begin{aligned} \mathrm{Cov}(\boldsymbol{Y}) &= \boldsymbol{C}\boldsymbol{\Sigma}\boldsymbol{C}^T \\ &= (\boldsymbol{C}\boldsymbol{L})(\boldsymbol{C}\boldsymbol{L})^T + \boldsymbol{C}\boldsymbol{\Psi}\boldsymbol{C}^T \\ &= \tilde{\boldsymbol{L}}\tilde{\boldsymbol{L}}^T + \tilde{\boldsymbol{\Psi}}. \end{aligned}$$

Consequences:

- Loadings rescale: $\tilde{\boldsymbol{L}} = \boldsymbol{C}\boldsymbol{L}$.
- Specific variances rescale: $\tilde{\psi}_j = c_j^2 \psi_j$.
- The factor-model structure is preserved under changes of units.

In practice: FA is often performed on the correlation matrix when variables have different units.

Rotational Invariance of Factor Analysis

Key idea: Different loading matrices can represent exactly the same covariance structure.

Let $\boldsymbol{R}$ be an orthogonal matrix:

$$\boldsymbol{R}\boldsymbol{R}^T = \boldsymbol{I}_m.$$

Define the rotated loadings and factors as

$$\boldsymbol{L}_* = \boldsymbol{L}\boldsymbol{R}, \qquad \boldsymbol{F}_* = \boldsymbol{R}^T\boldsymbol{F}.$$

Then the factor model is unchanged:

$$\boldsymbol{L}_*\boldsymbol{F}_* = \boldsymbol{L}\boldsymbol{R}\boldsymbol{R}^T\boldsymbol{F} = \boldsymbol{L}\boldsymbol{F}.$$

Similarly, the covariance structure is unchanged:

$$\boldsymbol{L}_*\boldsymbol{L}_*^T + \boldsymbol{\Psi} = \boldsymbol{L}\boldsymbol{R}\boldsymbol{R}^T\boldsymbol{L}^T + \boldsymbol{\Psi} = \boldsymbol{L}\boldsymbol{L}^T + \boldsymbol{\Psi}.$$

Why Rotate?

Rotation preserves model fit but can produce **more interpretable factor loadings**.

Rotation in Practice: Varimax

Goal: Obtain a simpler loading pattern without changing the fitted covariance matrix.

Before Rotation

	F1	F2
X_1	.55	.49
X_2	.62	.41
X_3	.48	-.55
X_4	.39	-.67

After Rotation

	F1	F2
X_1	**.75**	.10
X_2	**.72**	.15
X_3	.08	**-.72**
X_4	.12	**-.76**

- **Varimax** is a widely used orthogonal rotation.
- It seeks a **simple structure**: large loadings on a few variables and small loadings on others.

Key Takeaway

Rotation changes interpretation, not model fit.

Example: Common Factors in Stock Returns

We analyze weekly stock returns for **five companies** over 103 consecutive weeks.

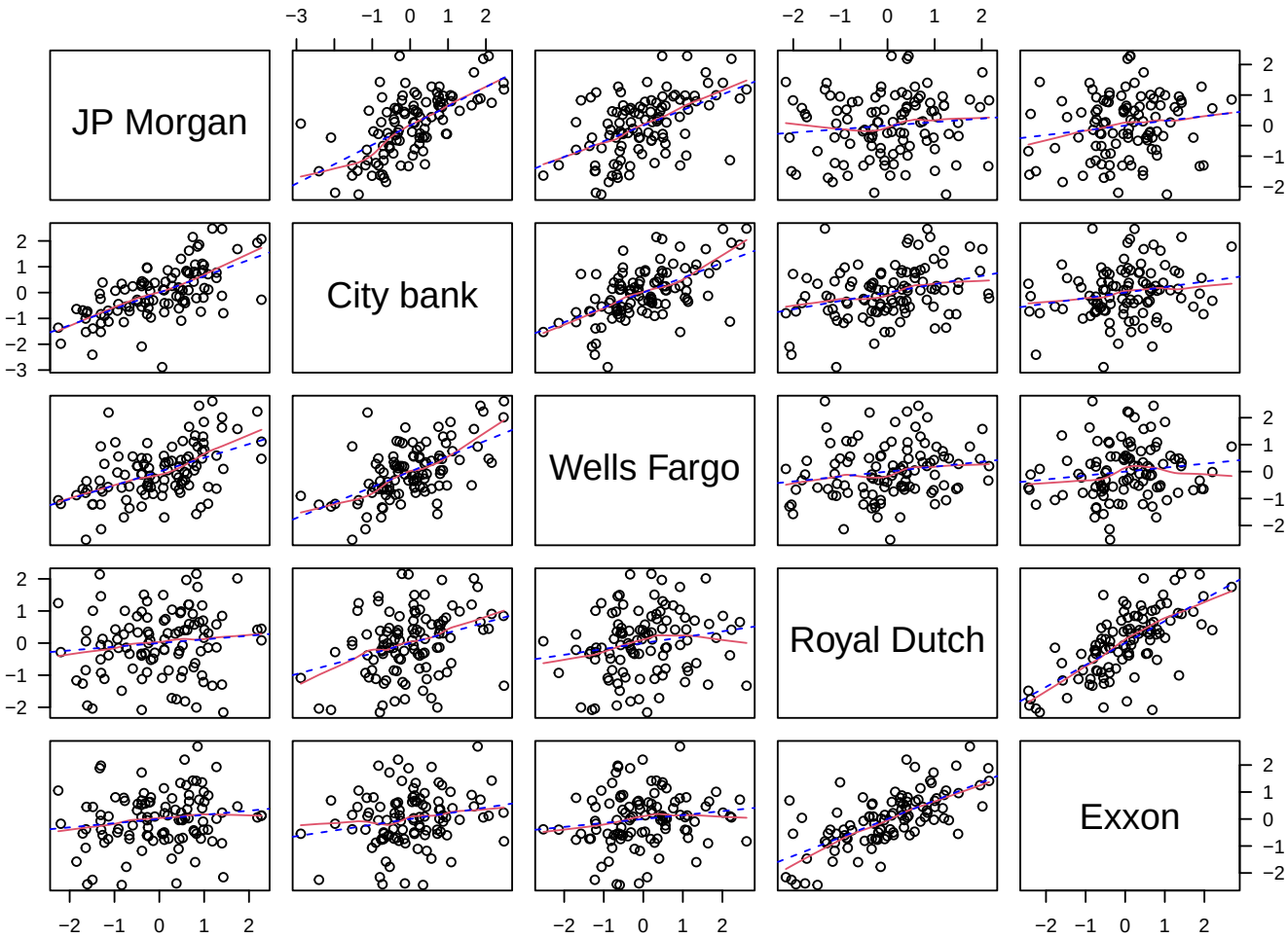
- **Banks:** JP Morgan, Citibank, and Wells Fargo.
- **Oil companies:** Royal Dutch Shell and Exxon.
- Returns are standardized, so we analyze the **sample correlation matrix**.

Research Question

Can the dependence among these five stocks be explained by a small number of **common market and sector factors**?

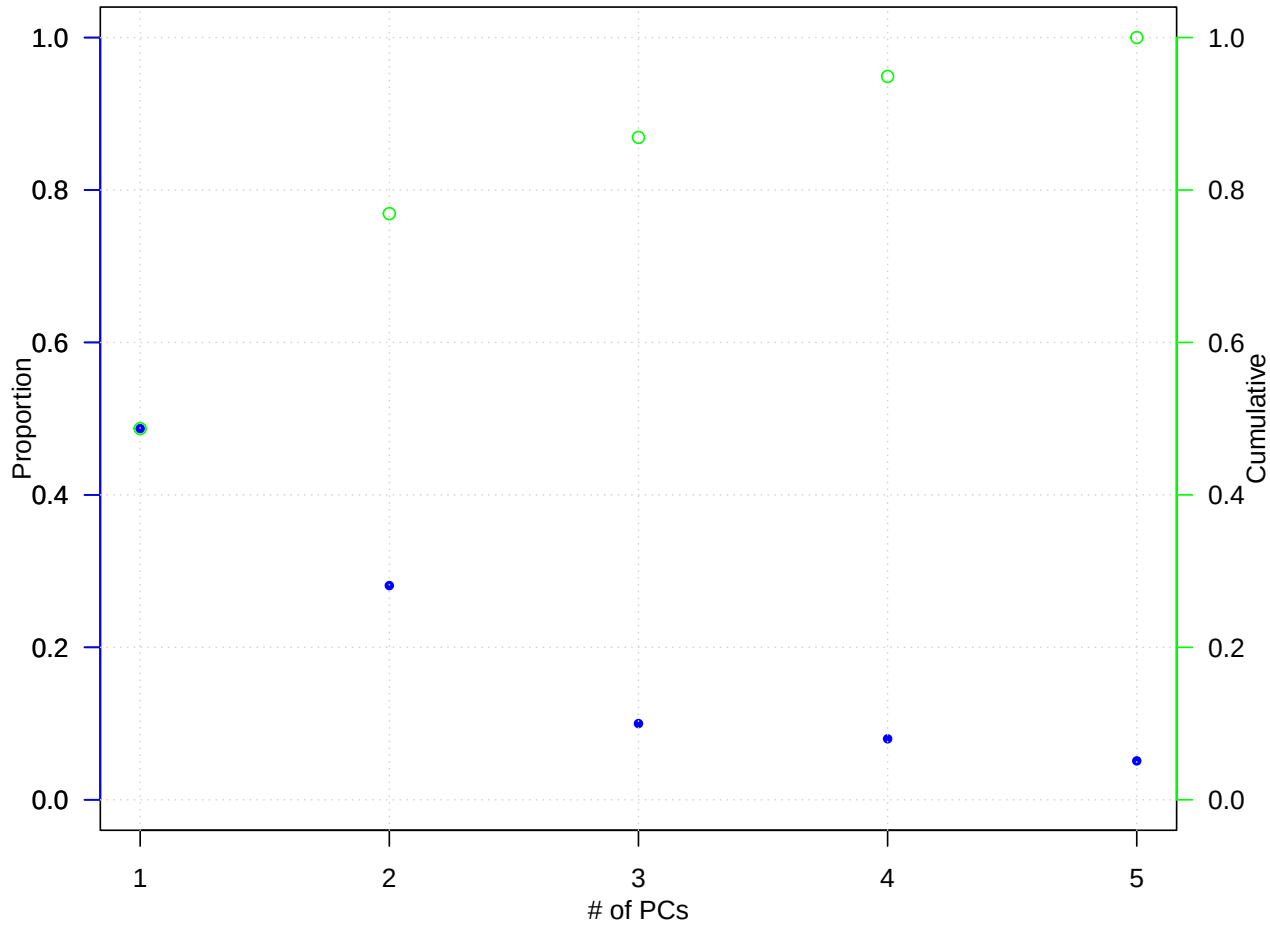
We begin by considering a two-factor model ($m = 2$).

Exploring Dependence Among Stock Returns



Question: Do stocks within the same industry appear more strongly correlated?

Choosing the Number of Factors



Question: Does the scree plot support retaining two common factors?

Recall: Eigenvalues provide exploratory guidance, not a definitive choice of m.

Stock Returns: Estimated Factor Loadings

Stock	Factor 1	Factor 2	Uniqueness
JP Morgan	0.732	0.437	0.273
Citibank	0.831	0.280	0.230
Wells Fargo	0.726	0.374	0.333
Royal Dutch	0.605	-0.694	0.153
Exxon	0.563	-0.719	0.166

- **Factor 1:** All loadings are positive $\Rightarrow$ possible overall market factor.
- **Factor 2:** Positive for banks, negative for oil stocks $\Rightarrow$ possible sector contrast.
- **Uniqueness:** Wells Fargo has the largest unexplained variance.

Does the Two-Factor Model Fit?

Compare the observed and fitted correlation matrices:

$$\boldsymbol{R}_{\text{res}} = \boldsymbol{R} - (\hat{\boldsymbol{L}}\hat{\boldsymbol{L}}^T + \hat{\boldsymbol{\Psi}}).$$

$$\begin{pmatrix} 0 & -.10 & -.18 & -.03 & .06 \\ -.10 & 0 & -.13 & .01 & -.05 \\ -.18 & -.13 & 0 & .00 & .01 \\ -.03 & .01 & .00 & 0 & -.16 \\ .06 & -.05 & .01 & -.16 & 0 \end{pmatrix}$$

Assessing Model Adequacy

Ideally, residual correlations should be close to zero.

Are residuals as large as -0.18 small enough to consider the fit adequate?

Maximum Likelihood Factor Analysis

Fit the two-factor model using maximum likelihood.

```
> (stock.fac <- factanal(stock, factors = 2,
+ method = "mle", scale = T, center = T))

Call:
factanal(x = stock, factors = 2, method = "mle", scale = T, center = T)

Uniquenesses:
  JP Morgan   City bank Wells Fargo Royal Dutch
      0.417       0.275       0.542       0.005
      Exxon
      0.530

Loadings:
            Factor1 Factor2
JP Morgan   0.763
City bank   0.819   0.232
Wells Fargo 0.668   0.108
Royal Dutch 0.113   0.991
Exxon       0.108   0.677

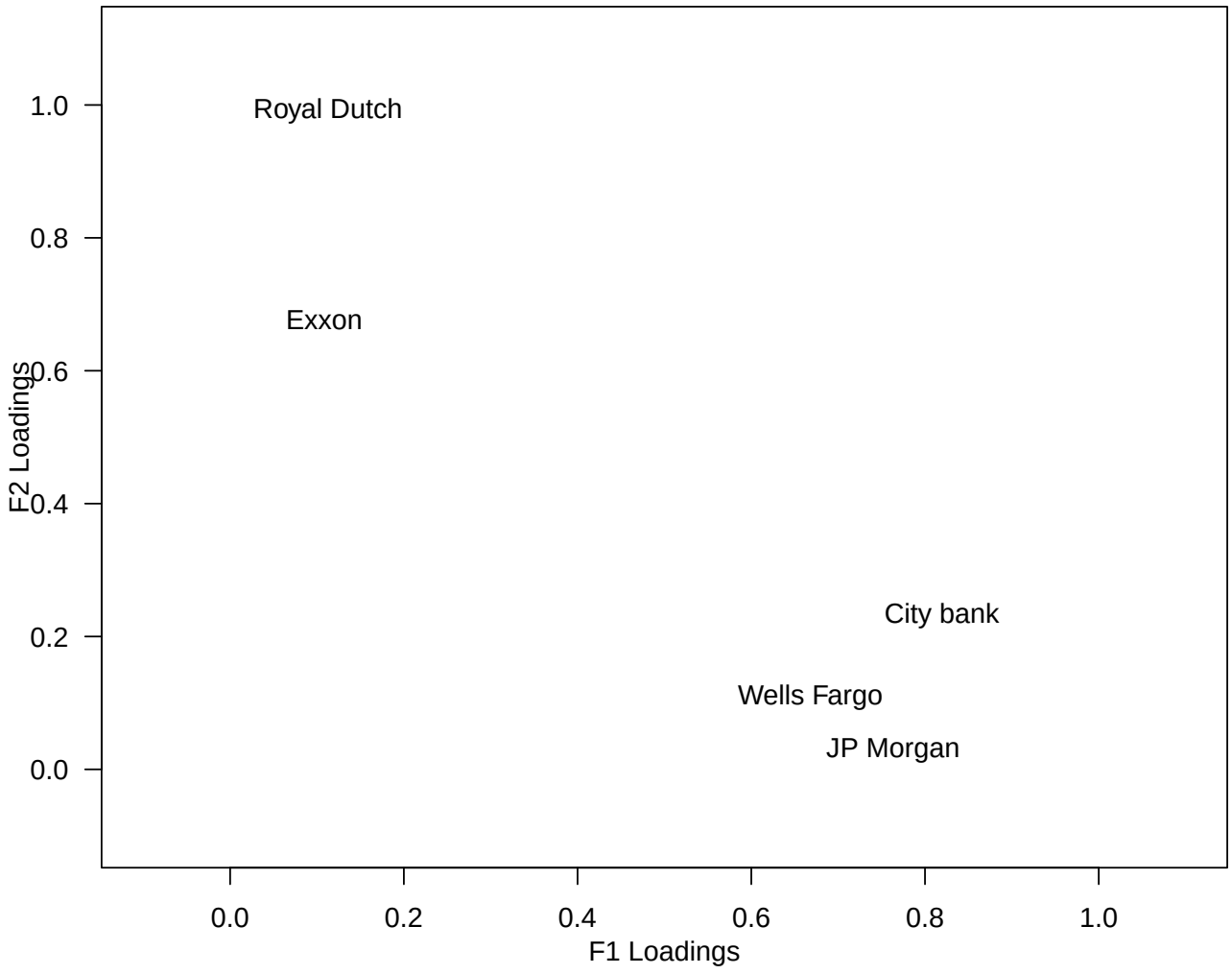
               Factor1 Factor2
SS loadings      1.725   1.507
Proportion Var   0.345   0.301
Cumulative Var   0.345   0.646

Test of the hypothesis that 2 factors are sufficient.
The chi square statistic is 1.97 on 1 degree of freedom.
The p-value is 0.16
```

Reading the R Output

Focus on the **estimated loadings**, **uniquenesses**, and **goodness-of-fit test**.

Interpreting the Two Common Factors



Interpretation: Do the loading patterns suggest an **overall market factor** and a **bank-versus-oil factor**?

Factor Analysis in Practice

1. **Explore dependence:** Examine correlations and assess suitability for FA.
2. **Choose candidate m:** Use scree plots and parallel analysis.
3. **Fit the model:** Estimate loadings and uniquenesses (e.g., ML).
4. **Assess model adequacy:** Examine residual correlations and goodness-of-fit.
5. **Rotate if useful:** Seek a simpler loading structure.
6. **Interpret factors:** Use loading patterns and subject-matter knowledge.

Goal

Explain observed covariance using a small number of **interpretable latent factors**.

PCA vs. Factor Analysis: Revisited

	PCA	FA
Goal	Summarize total variation	Explain covariance structure
Variables	Components constructed from $\boldsymbol{X}$	Unobserved common factors
Model	Algebraic decomposition	Statistical latent-variable model
Dependence on m	Leading PCs unchanged	Estimated factors may change
Rotation	PCs chosen to maximize variance	Loadings may be rotated for interpretation

Different Questions

PCA: How can we summarize $\boldsymbol{X}$?

FA: What latent structure may explain $\boldsymbol{X}$?

Factor Analysis: Key Takeaways

1. **Latent-variable model:** FA explains covariance using common factors.
2. **Covariance decomposition:** $\boxed{\boldsymbol{\Sigma} = \boldsymbol{L}\boldsymbol{L}^T + \boldsymbol{\Psi}}$
3. **Factor loadings:** Describe relationships between variables and factors.
4. **Communality and uniqueness:** Partition each variable's variance.
5. **Model selection and fit:** Choose m and assess model adequacy.
6. **Rotation:** Improves interpretability without changing model fit.

Remember

PCA summarizes variation; FA explains dependence through latent factors.